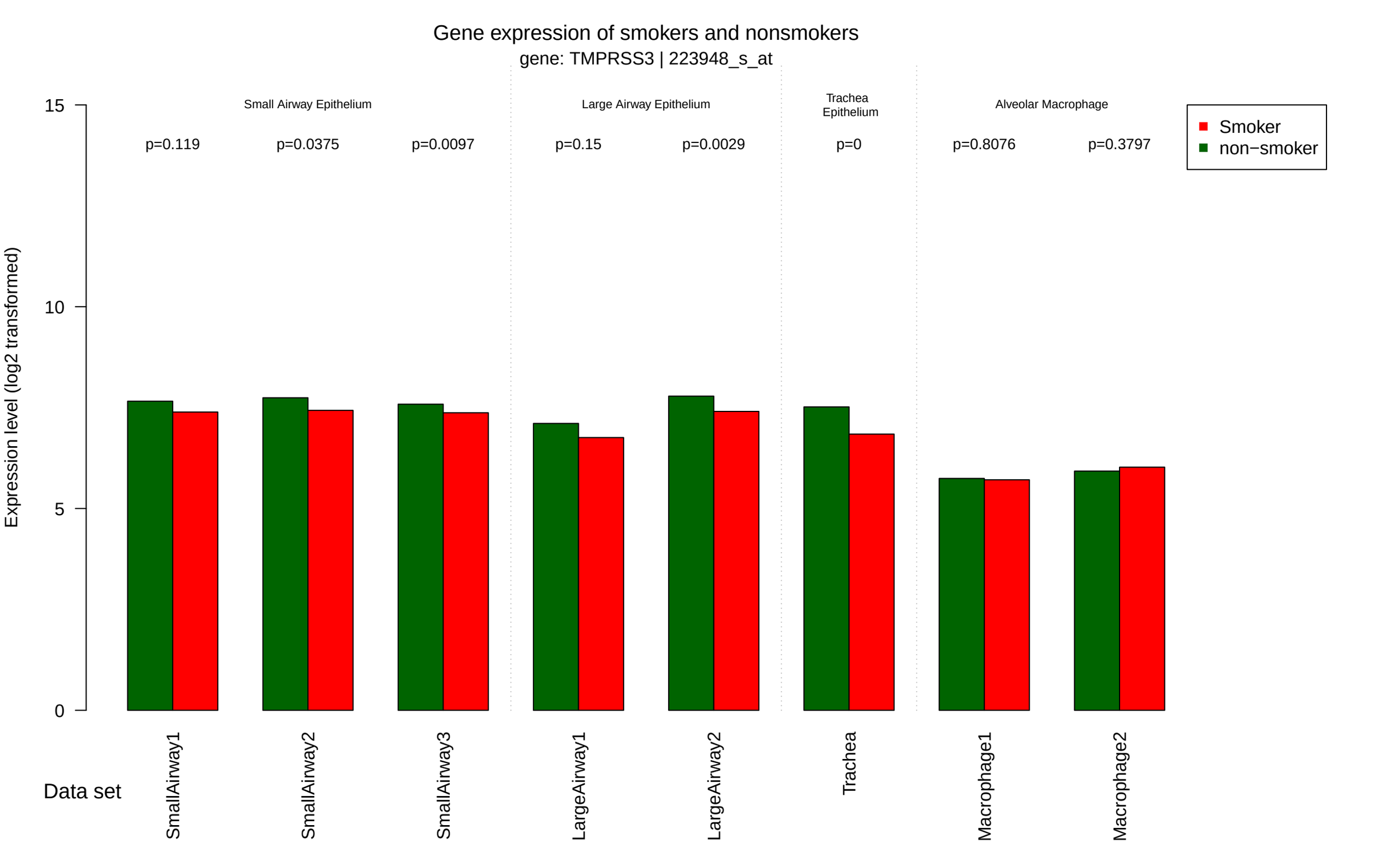
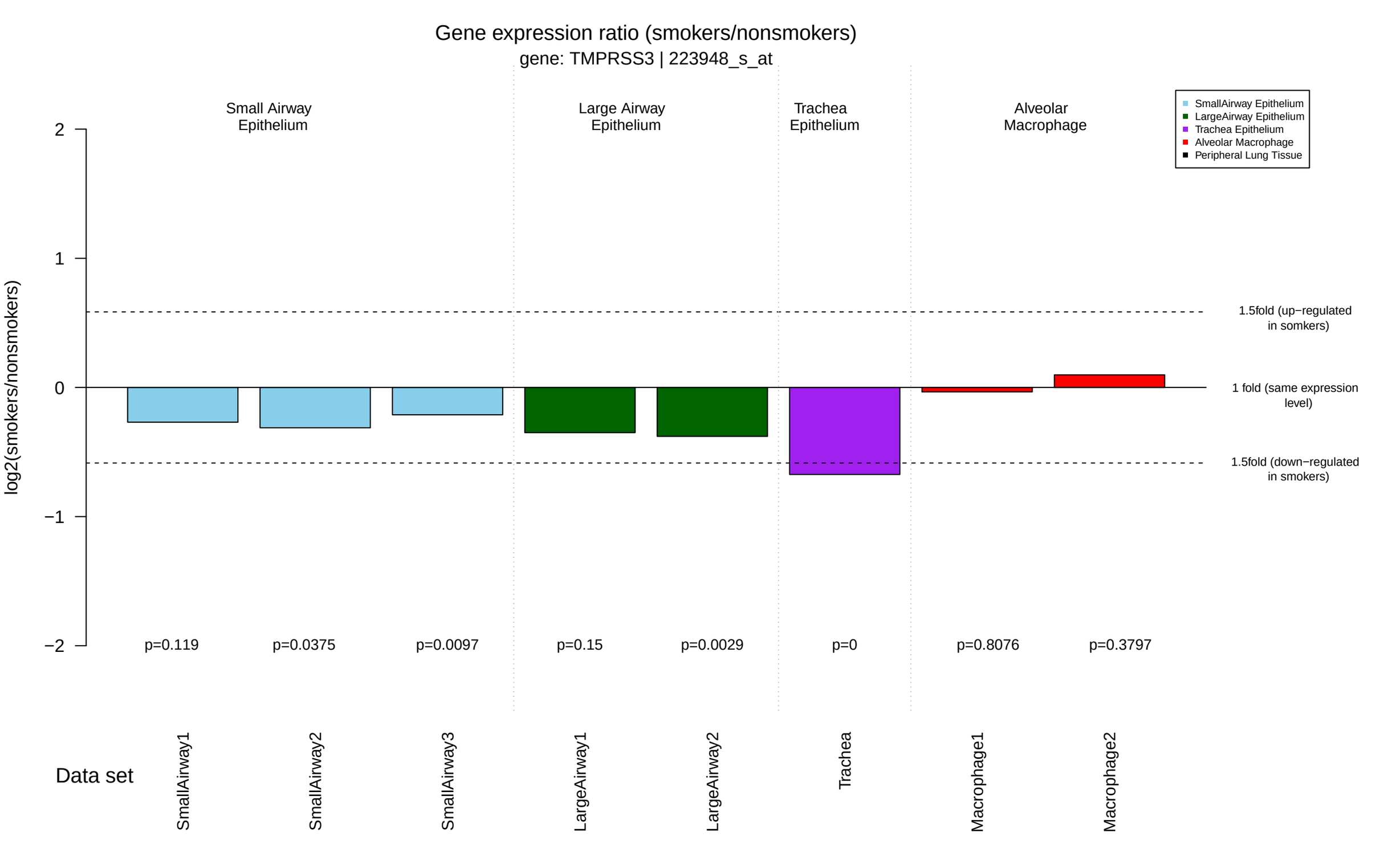
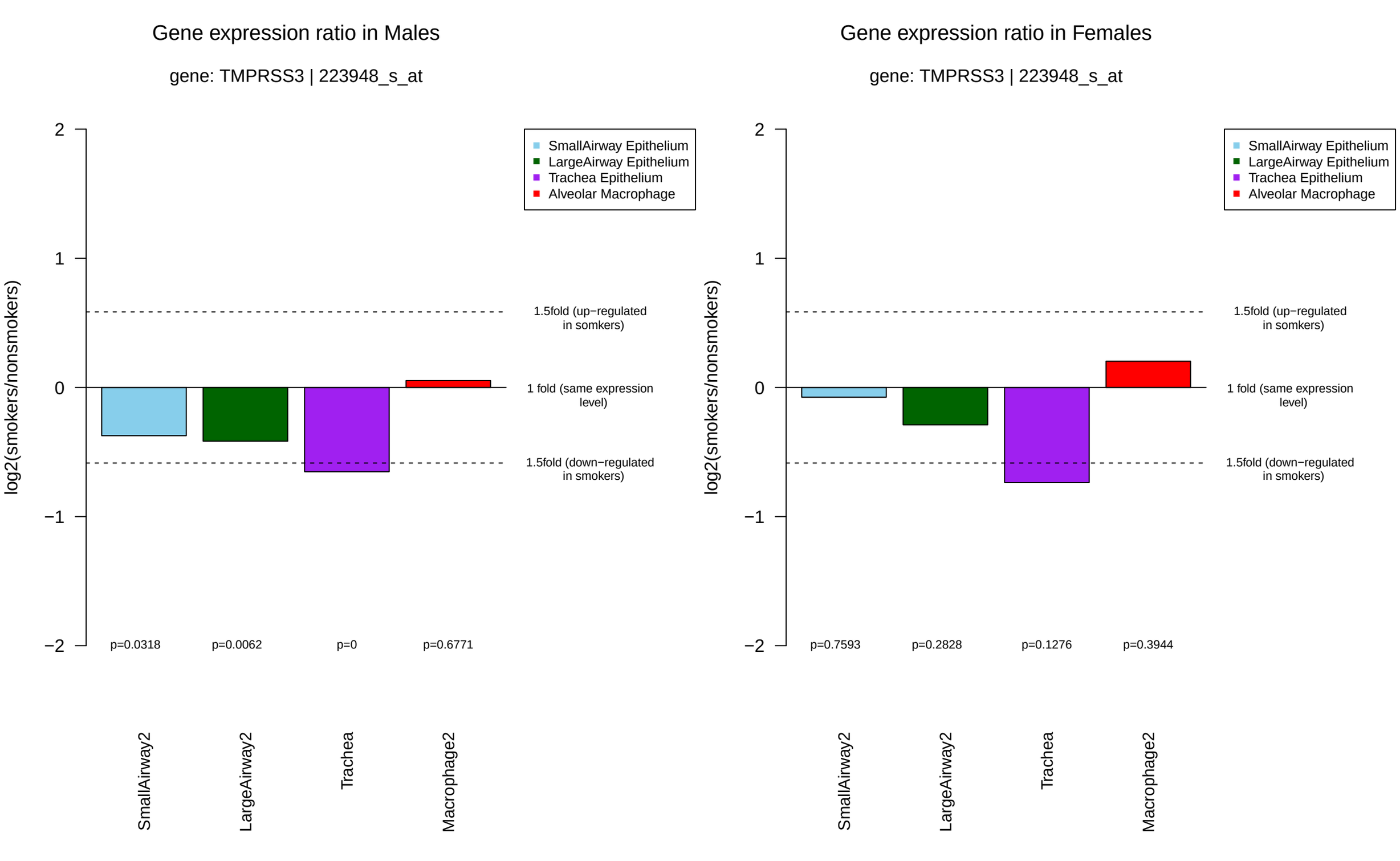
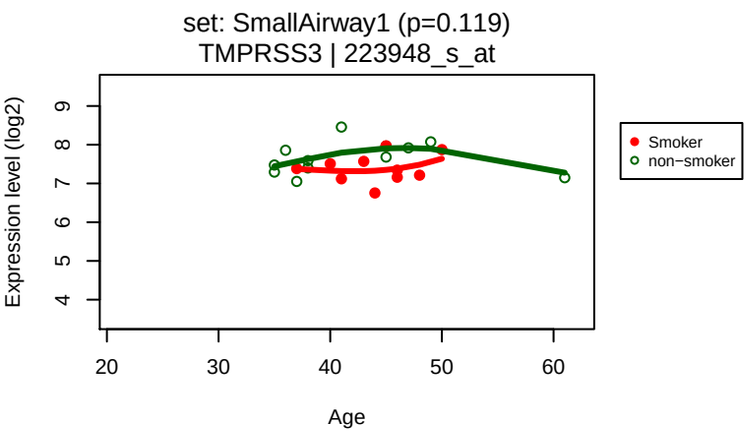
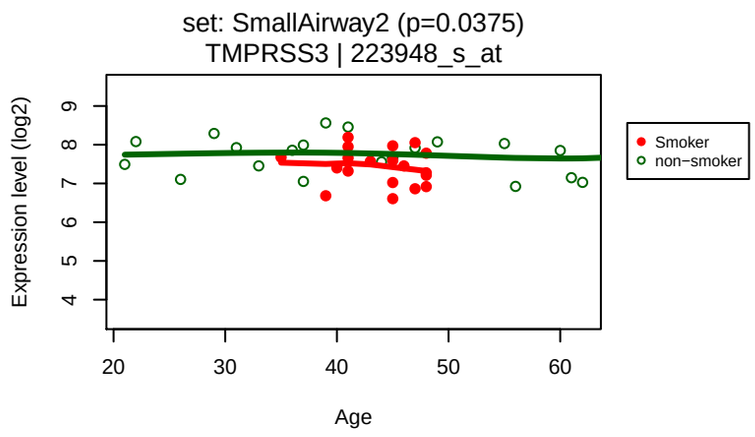
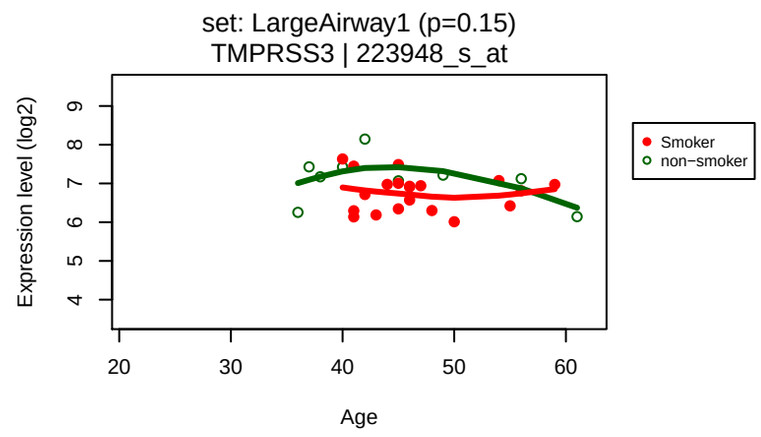

Gene expression of smokers and nonsmokers
gene: TMPRSS3 | 223948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.119
p=0.0375
p=0.0097
p=0.15
p=0.0029
p=0
p=0.8076
p=0.3797
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

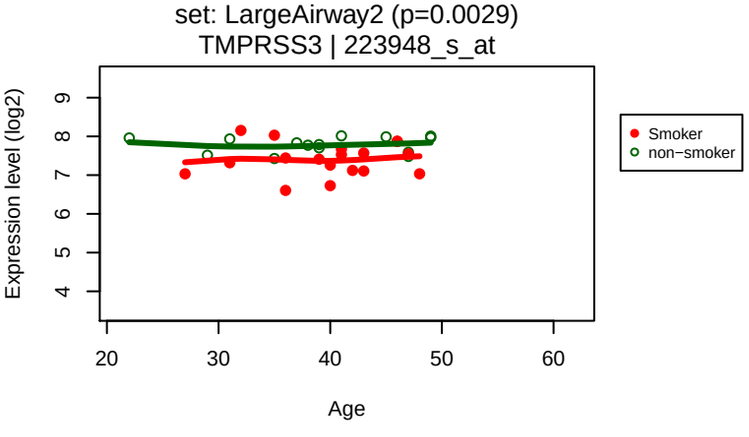

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS3 | 223948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.119
p=0.0375
p=0.0097
p=0.15
p=0.0029
p=0
p=0.8076
p=0.3797
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

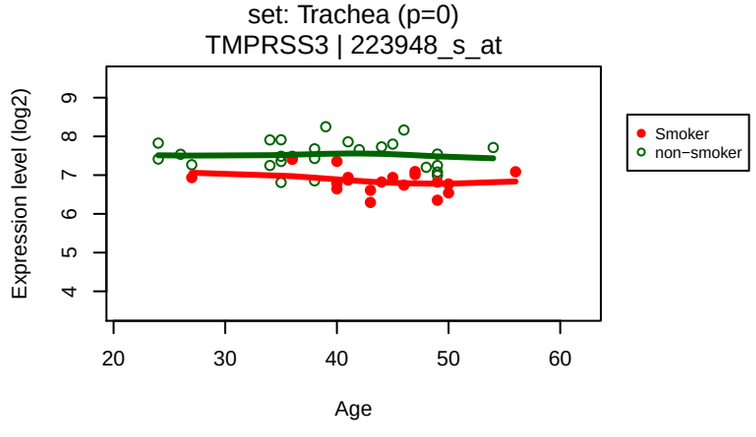

Gene expression ratio in Males
gene: TMPRSS3 | 223948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0318
p=0.0062
p=0
p=0.6771
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS3 | 223948_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7593
p=0.2828
p=0.1276
p=0.3944
SmallAirway2
LargeAirway2
Trachea
Macrophage2

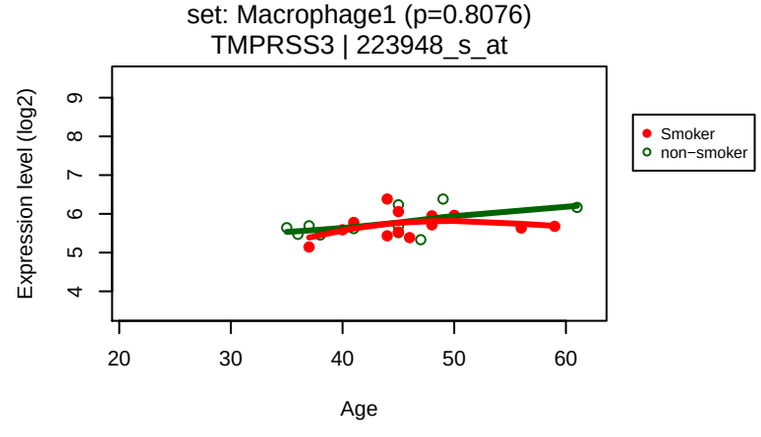

set: SmallAirway1 (p=0.119)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker

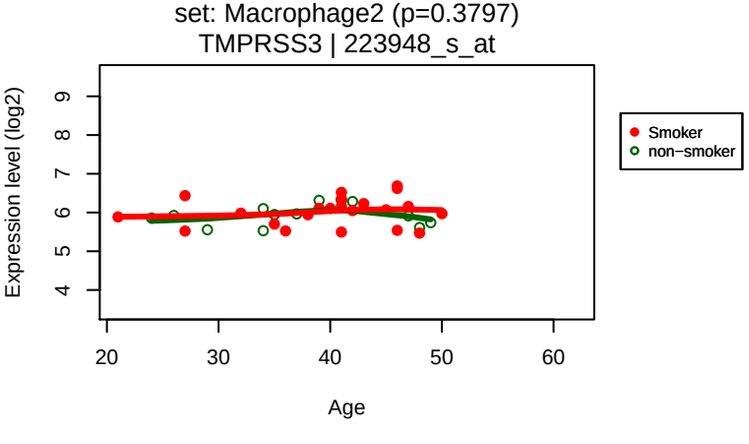

set: SmallAirway2 (p=0.0375)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker

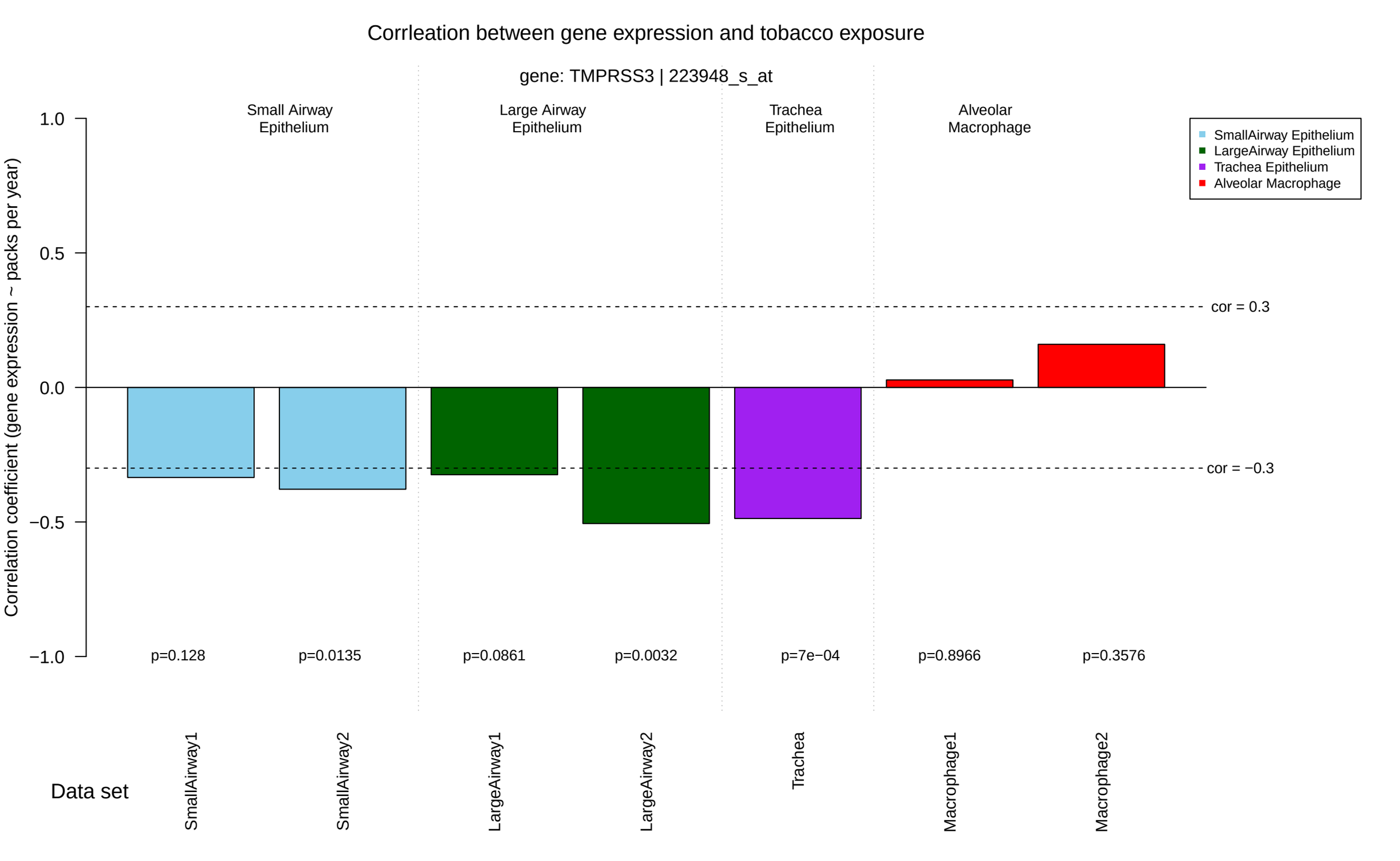

set: LargeAirway1 (p=0.15)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0029)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8076)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3797)
TMPRSS3 | 223948_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS3 | 223948_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.128
p=0.0135
p=0.0861
p=0.0032
p=7e−04
p=0.8966
p=0.3576
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2